



When herbal supplements cause heart problems: Clinical case report and forensic toxicological analysis

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ABSTRACT

The global expansion of travel and online supplement markets has eliminated traditional geographic barriers to toxic plant exposures. We report a case of a 42-year-old female presenting with digoxin-like intoxication symptoms, including nausea, vomiting, and hypotension. Electrocardiography revealed sinus bradycardia (51/min), first-degree AV block, and downsloping ST-segment depressions with reverse tick morphology. Upon further comprehensive anamnesis, the patient revealed the consumption of a weight-loss supplement labeled as "Tejocote root" (*Crataegus mexicana*), purchased during a travel to the USA. Serum digoxin assays yielded conflicting results across different platforms. The patient was managed with supportive care, including fluid replacement and antiemetics. Bradycardia persisted for five days, with complete resolution of symptoms and ECG abnormalities by day nine. Subsequent forensic toxicological analysis using liquid chromatography coupled to high-resolution mass spectrometry (LC-HRMS) and DNA barcoding confirmed the presence of yellow oleander (*Thevetia peruviana*) in the supplement, with no detection of the advertised *Crataegus mexicana*. While standard toxicological screening panels do not include plant-derived cardiac glycosides, cross-reactivity in digoxin assays may aid early diagnosis. Clinicians should maintain high suspicion for exotic plant poisonings in patients presenting cardiac glycoside-like symptoms, particularly following use of imported or online-purchased supplements. This case underscores the critical importance of multidisciplinary collaboration in determining the root cause of intoxication and highlights significant gaps in current forensic analytical capabilities for detecting botanical adulterants with public health implications.

1. Introduction

The global weight-loss supplement market reflects a widespread consumer demand for products marketed to achieve or maintain lean body weight [1]. From a clinical standpoint, these supplements purportedly function through various mechanisms: appetite suppression and reduced caloric intake, impaired nutrient absorption, enhanced fat oxidation, or laxative effects. Typical formulations contain dietary fiber,

caffeine, and minerals, often combined with botanical extracts or dried herbal material. While consumers often gravitate toward these botanical formulations under the misconception that "natural" ingredients are inherently safe, clinicians and toxicologists must remain vigilant about the significant health risks associated with these products. Multiple case reports consistently document contamination with prohibited pharmaceutical agents such as sibutramine, toxic concentrations of heavy metals or pesticides, and substitution with poisonous plant species

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[1–3]. While most adverse events present as gastrointestinal disturbances, severe toxicities requiring emergency medical intervention have been well-documented. A landmark case occurred in the early 1990s when Belgian physicians identified a cluster of rapidly progressive, irreversible renal disease among women attending a weight-loss clinic [4]. Investigation revealed inadvertent substitution of *Aristolochia fangchi* for a benign Chinese herb in the clinic's regimen. Similarly, in the United States, Plantain-labeled supplements contaminated with *Digitalis lanata* caused cardiotoxicity, with patients presenting elevated serum digoxin levels consistent with cardiac glycoside poisoning [5]. More recently, a fatal case in 2020 involved online-purchased candlenuts that were actually yellow oleander (*Thevetia peruviana*) seeds [6]. A case reported by a US poison center in 2022 prompted an FDA warning [7], and since 2023, the CDC and FDA have documented multiple cases where yellow oleander was substituted in supplements labeled as Mexican hawthorn (*Crataegus mexicana*) [8,9]. To our knowledge, no such reports have emerged at the European level. These three species—candlenut tree seeds ("Nuez de la India"), Mexican hawthorn root ("Raíz de tejocote"), and yellow oleander seeds ("codo de fraile")—represent popular weight loss products available along the US–Mexico border, despite being marketed with unsubstantiated safety and efficacy claims [10]. Notably, despite the documented cases in North America, to our knowledge, no intoxication cases involving these botanical supplements—particularly tejocote or yellow oleander adulteration—have been reported at either the European Union level or, at least to our knowledge, at the national level in the different individual European countries. Here we present the first documented clinical case of yellow oleander intoxication in Europe involving weight-loss supplement contamination. This study aimed to: (1) characterize the clinical presentation and management of cardiac glycoside poisoning from an adulterated tejocote supplement in a European healthcare setting; (2) identify the cause of this event by analyzing the supplement by means of liquid chromatography coupled to tandem high-resolution mass spectrometry (LC-HRMS/MS or LC-HRMS^E) and DNA barcoding; and (3) highlight the diagnostic challenges and public health implications of plant-based supplement adulteration in regions where such poisonings are uncommon.

2. Material and methods

2.1. Status of consent

In line with the COPE best practice guidelines, written consent of the patient for publication was obtained.

2.2. Serum digoxin quantifications

According to current European Society of Cardiology guidelines, monitoring serum digoxin concentrations is recommended to avoid toxicity in patients treated with digoxin. Due to suspected digoxin toxicity, a serum quantification assay was performed. On the admission day, both an electrochemiluminescence immunoassay or ECLIA (Elecsys Digoxin®, Roche) and a Kinetic Interaction of Microparticles in Solution (KIMS) immunoassay (ONLINE TDM Digoxin®, Roche) were used to determine serum digoxin levels. While the KIMS assay delivers faster results, the ECLIA assay offers greater robustness and sensitivity. Specifically, the ECLIA has a lower theoretical limit of detection (LOD) of 0.2 ng/mL (0.27 nmol/L) compared to 0.3 ng/mL (0.38 nmol/L) for KIMS. However, both assays share the same lower limit of quantification (LLOQ) of 0.4 ng/mL (0.51 nmol/L).

2.3. Forensic toxicological analysis of the supplement

The supplement underwent comprehensive analysis using both chemical characterization via liquid chromatography coupled to high-resolution mass spectrometry (LC-HRMS/MS) and molecular

identification through DNA barcoding. Chemical analysis was conducted at two independent official medicines control laboratories routinely involved in the analysis of medicines and health products, including suspicious herbal supplements: Swissmedic and Sciensano. Sciensano performed qualitative analysis using ultra-high-performance liquid chromatography (UHPLC) with quadrupole-Orbitrap mass spectrometry (LC-HRMS/MS), while Swissmedic conducted both qualitative analysis and quantitative estimation using UHPLC equipped with a diode array detector (DAD), and coupled to quadrupole time-of-flight mass spectrometry (LC-HRMS^E). Identification testing was performed in compliance with the requirements of ISO/IEC 17025 and the General European OMCL Network Quality Management Guidelines [11]. Toxins were reported as present only upon independent confirmation by the use of a reference standard and when detected by both laboratories. Technical details of the analytical methods, the acceptance criteria for a proper identification, and the determination of the limit of detection of the different cardiac glycosides are provided in the [supplemental data](#).

For molecular identification, DNA barcoding targeted two genetic loci: the species-specific Internal Transcribed Spacer 2 (ITS2) region and the more general but less discriminative ribulose 1–5 biphosphate carboxylase Large subunit (rbcL) locus. These loci were chosen for their proven success for many vascular plants, availability of universal primers, and extensive curated reference databases [12–17]. While the plastid rpoB locus was used previously and successfully distinguished between *Thevetia peruviana* and *Crataegus* sp. [18], ITS2 and rbcL were preferred as they are the standard markers for plant (meta)barcoding. The complete technical details regarding DNA extraction methodology, primers, amplification conditions, and sequencing protocols are described in the [supplemental data](#). Two DNA barcode identification approaches were employed following established methodologies [19, 20]. Both methods rely on BLAST similarity-based identification of sequences available in GenBank [21] and involve either considering only the top hits with maximum scores or applying a similarity method that assigns additional weight to the identity value in query-reference comparisons. DNA sequences were queried using megaBLAST through NCBI nucleotide BLAST [22,23]. The Top Hit Method considered hits within 10 points of maximum score with $\geq 70\%$ query coverage and $\geq 98\%$ identity. The Similarity Approach calculated scores for up to 100 BLAST hits using: maximum score $\times$ (query coverage/identity). Taxonomic assignments followed hierarchical confidence levels: species-level required single species identification with $\geq 98\%$ identity; genus-level for multiple species from same genus; family-level for multiple genera within one family, all maintaining $\geq 98\%$ identity thresholds. Reference material for Mexican Hawthorne and Yellow oleander, obtained from different botanical gardens, were used to demonstrate the applicability of the used methodology.

3. Results

3.1. Clinical observations

A 42-year-old female presented to the emergency department (ED) with persistent nausea and vomiting starting the previous night. Her symptoms remained refractory to metoclopramide and metamizole administered by her primary care physician. She denied experiencing diarrhea or fever. Nine months prior, she had undergone a gastric bypass for obesity, with no subsequent complications. Her current medications were limited to pantoprazole and vitamin supplements. A chronological description of the symptoms and treatment is listed in [Table 1](#).

Clinical examination revealed mild hypotension (104/41 mmHg) and bradycardia (51/min). Although afebrile, the patient appeared lethargic and dehydrated. Apart from diminished bowel sounds, the abdominal and cardiopulmonary examinations were normal. Laboratory tests showed leukocytosis (14.0 G/L) with neutrophilia and monocytosis, along with mildly elevated alkaline phosphatase (110 U/L). Electrolytes, liver function tests, and cardiac biomarkers remained

Table 1
chronological overview of the clinical symptoms and treatments given.

Day 1 & 2	Ingestion of 2 × 2 capsules (total of 4) of Tejocote contaminated with yellow oleander.
Day 3	Sudden onset of nausea and vomiting, with no improvement after metoclopramide and metamizole administered by primary care physician.
Day 4	Presentation in our emergency department with sinus bradycardia, first-degree AV block and ST-segment depression in ECG, yellow oleander intoxication suspected.
Day 5 & 6	Sinus bradycardia and first-degree AV block intermittently persisting, nausea and vomiting regressing.
Day 7	Patient discharged in good condition; sinus bradycardia and AV block resolved; ST-segment changes still visible.
Day 9	Follow-up: Normalized ECG, no clinical symptoms.

within normal limits. A contrast-enhanced CT scan of the abdomen showed a normal postoperative situs. The ECG revealed sinus bradycardia, new-onset first-degree AV block (PQ-time 226 ms), and downsloping ST-segment depressions with a reverse tick morphology (Fig. 1A). These findings prompted immediate digoxin level measurement, which yielded conflicting results: 0.38 nmol/L (0.297 µg/L) with the ECLIA test and 0.5 nmol/L, (0.39 µg/L) using the KIMS assay. In the case of the ECLIA assay, the obtained value is above the LOD, but below the LLOQ, while for the KIMS test, the obtained value corresponds to the LLOQ. Nevertheless, these values are well below the 1 µg/L serum levels that have been associated with digoxin toxicity [24]. However, it also has to be stated that serum digoxin levels will vary depending on the timing of the test and the patient's response to the toxin.

Initially, the patient denied ingesting other medications. However, while obtaining a more detailed history, she disclosed consuming four capsules of a weight loss supplement purchased in the USA containing “Tejocote root” (*Crataegus mexicana*). Subsequent online investigation revealed a recent FDA warning regarding the contamination of this product with yellow oleander or *Thevetia peruviana* [7–9]. Based on this information, yellow oleander intoxication was suspected, and the patient was admitted for further monitoring and supportive therapy with metoclopramide, ondansetron, and Ringer-Acetate solution.

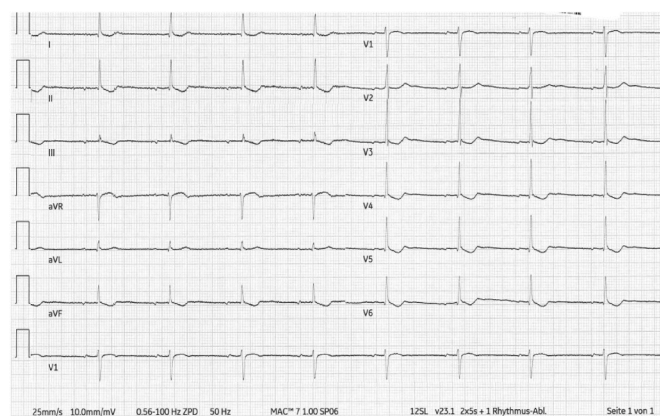
During the observation period, sinus bradycardia with heart rates as low as 31/min occurred. Administration of digoxin immune Fab was considered but not indicated as repeat measurements of serum potassium remained less than 5.0 mEq/L, and no arrhythmias or hemodynamic instability developed during the observation period. As the ingestion occurred more than 48 h prior to the ED presentation, no gastric decontamination was performed. The continuous cardiac monitor showed no high-grade AV blocks or arrhythmias throughout the

admission. Sinus bradycardia, ST-Segment changes, and first-degree AV block persisted until the fifth day after the supplement's ingestion, with serum potassium levels remaining within the normal range. The patient was discharged asymptomatic and in good condition. ST-segment changes were completely resolved on the follow-up ECG after nine days (Fig. 1).

3.2. Forensic chemical and molecular analysis of the herbal supplement

The herbal supplement underwent comprehensive screening for cardiac glycosides using liquid chromatography coupled with high-resolution tandem mass spectrometry (LC-HRMS:MS (Sciensano) or LC-HRMS^E (Swissmedic), with detailed methodology provided in the supplemental data. Chemical analysis results are presented in Fig. 2 and Table 2, confirming the presence of four cardiac glycosides: Thevetin A, Thevetin B, cerberin, and neriifolin—a combination characteristically found in plants from the *Thevetia* genus [25,26]. Subsequent estimation of the amount of Thevetins present indicated that approximately 28 mg Thevetin B and 3.1 mg of Thevetin A were present per gram of sample. However, due to the wide range of the mass of the supplement “chunks” (see supplemental material and methods) and material heterogeneity, it is impossible to determine the exact amount of these toxins ingested. Moreover, additional cardiac glycosides typically associated with *Nerium oleander* or foxglove species were found absent in the sample (Table 2), effectively excluding these plants as potential sources of the observed clinical toxicity. However, due to the inherent chemical complexity of herbal material, the sample was independently assessed by DNA barcoding, targeting both the nuclear internal transcribed spacer 2 (ITS2) locus and portions of the *rbcLa* gene (see detailed methodology in the supplemental data). Both top hit and similarity approaches were employed for taxonomic identification as shown in the supplemental data [19,20]. To determine which genetic marker was most suitable for plant identification, we first tested reference material using both the *rbcLa* and ITS2 loci. This analysis showed that the *rbcLa* locus has limited discriminative power, making it less suitable for identifying plant species in this type of study (detailed results are provided in the supplemental data). In contrast, the ITS2 marker successfully identified plant material from the *Thevetia* genus, specifically *Thevetia peruviana* (yellow oleander). When the herbal supplements were tested using this marker, they were found to contain *Thevetia peruviana* with no other plant species detected. This confirmed that the supplement consisted mainly of this toxic plant species. The convergent results from both chemical and molecular analyses provided robust evidence that the clinical presentation resulted from *Thevetia peruviana* contamination rather than the advertised *Crataegus mexicana* content.

Upon admission:



7 days later follow-up:

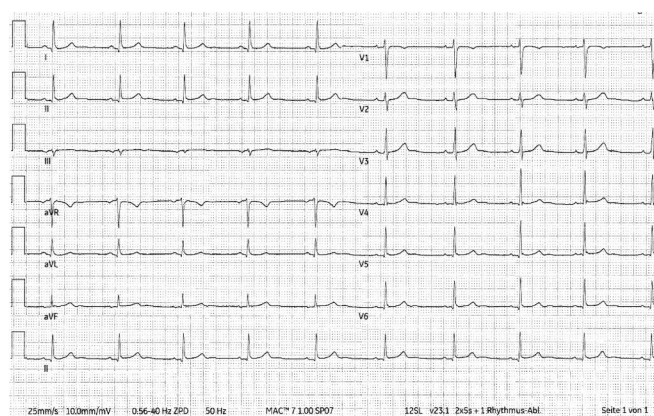


Fig. 1. Left panel demonstrates the ECG on admission 48 h after the last dose. Evidence of sinus bradycardia, first-degree AV block, and downsloping ST-segment depressions with a reverse tick, while the right panel displays the ECG follow-up 9 days after ingestion. Initially observed changes were completely resolved.

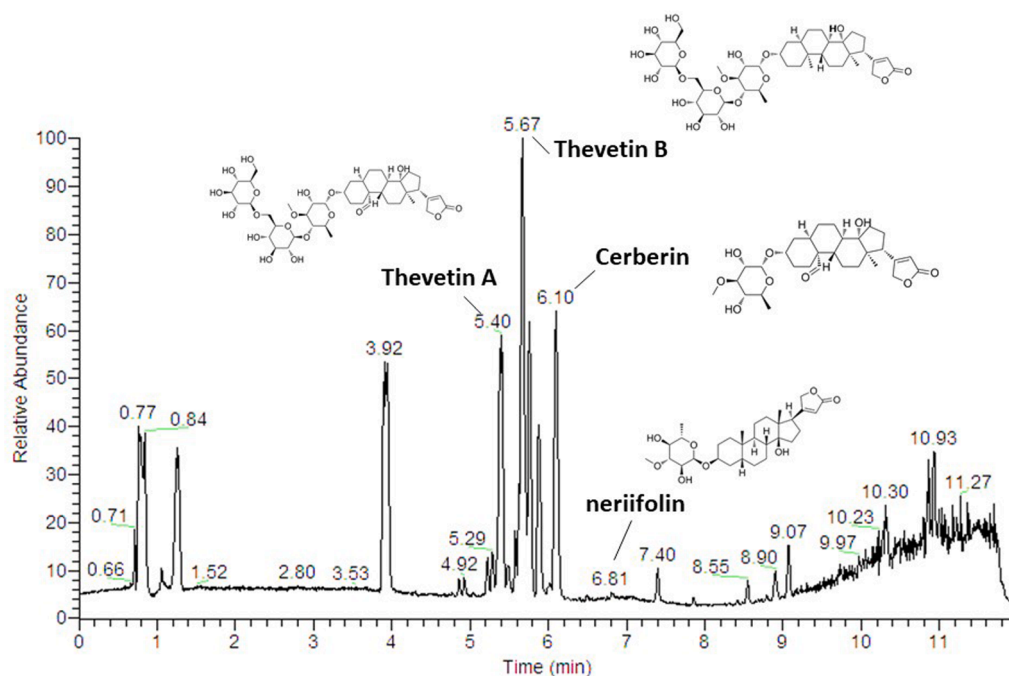


Fig. 2. Total ion chromatogram of the gummie extract and the annotation of the different cardiac glycosides detected in the sample. The other cardiac glycosides were either not present in the samples or were present in amounts below the limit of detection (LOD) of the employed methodology.

Table 2

Overview of the outcome of the chemical analysis and DNA barcoding of the sample. Both the more specific internal transcribed spacer region 2 (ITS2) and the more general *rbcl* locus were amplified and analysed.

Found in the sample		
Cardiac glycosides	Digoxin	-
	Digitoxin	-
	Digitoxigenin	-
	Oleandrin	-
	Thevetin A	+
	Thevetin B	+
	Cerberin	+
	Peruvoside	-
	Neriifolin	+
ITS2 as barcode region	Top hit	<i>Thevetia peruviana</i>
	Similarity approach	<i>Thevetia peruviana</i>
<i>rbcl</i> as barcode region	Top hit	<i>Thevetia</i> sp.
	Similarity approach	<i>Thevetia</i> sp.

4. Discussion

We present a patient with characteristic clinical symptoms of digoxin-like poisoning after consuming a weight-loss herbal supplement. Chemical analysis reveals the presence of cardiac glycosides, including Thevetin A, Thevetin B, cerberin, and neriifolin, which are marker molecules for *Thevetia peruviana* (yellow oleander) and were reported absent in *Crateagus mexicana* [27]. Additionally, cardiac glycosides commonly linked to *Nerium oleander* or foxglove species [28] were not found in the sample (Table 2), ruling out these plants as sources of the observed toxicity. The absence of oleandrin in this sample and in reference material of *Thevetia peruviana* contradicts a highly cited and authoritative reference work on *Thevetia* [29] but concurs with a more recent finding where oleandrin were absent in leaf samples from yellow oleander [30]. This discrepancy could be due to oleandrin levels being below our detection limit or because the compound previously identified as oleandrin was actually cerberin, as they share the same chemical formula and precursor molecule (see supplemental data). However, comparison with commercially available standards of oleandrin and

cerberin shows that these molecules differ in retention time and produce distinct MS² fragment ions (see supplemental data), demonstrating the importance of these verifications and the use of MS/MS fragmentation. Additional DNA barcoding analysis confirmed the chemical data, demonstrating the substitution of Hawthorne with yellow oleander in the supplement.

This substitution may have also occurred in previous clinical reports of intoxication with weight loss supplements claiming to contain *Crateagus* [31–35]. Based on the clinical findings, it was assumed that *Crateagus* itself was the source of this digoxin-like intoxication [31–35]. However, no forensic analysis was performed on the supplements to verify the presence of cardiac glycosides or digoxin or plant species that were actually present in the supplement. These cases underscore a critical need: when supplement-related intoxication is suspected, comprehensive forensic analysis must be conducted to unambiguously identify the causative agent [7]. Without such rigorous investigation through multidisciplinary collaboration, there is a real risk of wrongfully attributing toxicity to innocent plant species and failing to identify the actual causative agent.

Yellow oleander toxicity presents with a clinical picture similar to digoxin poisoning, as the abovementioned cardiac glycosides, inhibit the Na⁺-K⁺-ATPase, leading to hyperkalemia and increased intracellular calcium [29]. Additionally, cardiac glycosides increase the vagal tone, resulting in positive inotropic effects combined with negative chronotropic and dromotropic effects. Clinical manifestations of acute poisoning typically begin with gastrointestinal symptoms such as nausea and vomiting, followed by cardiac effects including bradycardia and impaired conduction, heart block, ventricular ectopy, and severe arrhythmias, while possible neurological symptoms such as confusion, drowsiness could occur. Hyperkalemia is common in yellow oleander poisoning, and may be an important marker for toxicity [28,29,36], but normal serum potassium and severe cardiac toxicity is documented [35]. Cardiac glycosides of *Thevetia* may interfere with digoxin immunoassays and are reported to result in apparent serum digoxin concentrations [37, 38]. This cross-reactivity could explain the conflicting results by different immunoassays in our case.

In our case, the ingestion of the substituted herbal supplement

resulted in gastrointestinal distress, sinus bradycardia, and AV block. Interestingly, the patient did not develop hyperkalemia, likely due to potassium loss from recurrent vomiting. Treatment of yellow oleander poisoning depends on the occurrence of cardiac arrhythmias and potassium level [36]. Milder cases can be handled with supportive therapy, including fluid replacement and antiemetics with close cardiac and electrolyte monitoring. Bradyarrhythmia can be treated with atropine, isoprenaline, and temporary pacing if necessary. Treatment of ventricular tachyarrhythmias with lidocaine is recommended, whereas the use of amiodarone and calcium channel blockers is contraindicated. Digoxin-Fab antibody fragments have a high binding affinity to digoxin and are demonstrated as an effective intervention in acute digoxin poisoning as well as in yellow oleander poisoning [39,40]. Indications may be prolonged gastrointestinal symptoms, but in cases with severe hyperkalemia or life-threatening arrhythmias digoxin-Fab antibody fragments are the treatment of choice [28,36,37,39]. Dosing Fab antibody fragments in yellow oleander poisoning is not established, but reportedly was effective with higher doses than for digoxin poisoning, due to different binding affinities and pharmacokinetic properties [28, 41]. Although the administration of magnesium and calcium is discussed, it is not routinely recommended and should particularly be avoided in hyperkalemic patients [36]. Cardioversion should be avoided as it is often unsuccessful and bears a risk of degeneration into ventricular fibrillation or the occurrence of asystole [28,36]. Hemodialysis or hemofiltration has no benefit for the elimination of glycosides but may be considered in severe hyperkalemia [36]. Gastric decontamination using activated charcoal within two hours of ingestion, or in some cases multiple-dose activated charcoal administration to interrupt the enterohepatic circulation, may be considered, especially if no digoxin-Fab antibody fragments are available [28,42]. Blood or urine analysis for digoxin might be positive due to a cross-reaction, but a negative result does not rule out cardiac glycoside poisoning [41,43].

Due to broad variability in clinical courses, predicting clinical prognosis might be difficult. The onset and duration of the symptoms depend on different factors, such as the varying toxin content of the plant components and their dosage form. In some cases, arrhythmias were reported up to 92 h post-ingestion [29,36,43]. In our case, delayed absorption due to the gastric bypass, as well as impaired uptake of the dried plant components, might have played a role. Timely recognition of botanical adulteration is clinically crucial, as digoxin-specific fragment antigen-binding (Fab) antibodies represent a targeted treatment option that should be promptly considered in any patient presenting with typical symptoms of cardiac glycoside poisoning. Importantly, identifying the specific botanical adulterant may be pivotal in guiding treatment decisions, as the required therapeutic approach and dosing can vary considerably depending on the causative agent.

5. Conclusion

This case represents the first documented instance of yellow oleander intoxication in Europe resulting from adulterated weight-loss supplements and addresses critical gaps in European clinical and forensic toxicology practice.

Regarding the clinical presentation and management: this case demonstrates the importance of obtaining thorough patient histories and including over-the-counter supplements in diagnostic workups. In patients presenting with gastrointestinal symptoms, somnolence, and ECG abnormalities, specifically bradycardia, cardiac glycoside poisoning—including *Thevetia* or *Nerium* species—should be considered in the differential diagnosis. Clinicians should recognize that plant-based intoxications may yield conflicting digoxin assay results due to cross-reactivity between structurally similar cardiac glycosides, potentially leading to false-positive or inconsistent results across different platforms. Additionally, Fab fragment therapy for severe plant-based cardiac glycoside intoxications may require higher doses than traditional digoxin poisoning and should be considered in a patient presenting with

typical clinical symptoms.

From an analytical methodology perspective, this case validates the effectiveness of combining LC-HRMS/MS with DNA barcoding as a robust framework for definitive botanical identification in supplement adulteration cases. The successful detection of yellow oleander-specific cardiac glycosides (Thevetin B, Thevetin A, neriifolin and cerberin) alongside molecular confirmation of *Thevetia peruviana* demonstrates that this integrated approach can overcome the limitations of either technique used in isolation. Importantly, this case underscores the need for comprehensive analytical approaches in European forensic toxicology laboratories, as these cardiac glycosides are absent from many standard European toxicological screening protocols for both biological matrices (e.g. blood or serum) and botanical supplement analysis [43–45,46]. Incorporating these compounds into routine analytical methods would significantly benefit forensic laboratories, especially as global travel and online marketplaces have eliminated traditional exposure boundaries.

Regarding public health implications and diagnostic challenges, this case highlights the availability of adulterated botanical supplements to European consumers through online retailers, as several of the FDA-listed brands were available to the European public. Moreover, this study underscores the importance of multidisciplinary collaboration among clinicians, poison centers, and specialized forensic testing laboratories for accurate diagnosis. Without rigorous analytical investigation, the actual toxic agent may remain undetected, incorrect safety profiles become established in the literature, and public health risks persist unrecognized. Furthermore, this case advocates for stronger national regulatory oversight of herbal supplements in the EU — whether sold in retail stores, online, or through social media — to ensure their safety and authenticity. In the USA particularly, the public should be warned about the risks of purchasing herbal supplements that lack independent third-party testing.

CRedit authorship contribution statement

Celine Vanhee: Writing – original draft, Visualization, Validation, Supervision, Resources, Investigation, Data curation, Conceptualization. **Theresa Theuer:** Writing – original draft. **Katrin Faber:** Writing – review & editing, Writing – original draft, Supervision. **Roosens Nancy H. C.:** Writing – review & editing, Supervision, Resources, Project administration. **Adrian Kindler:** Investigation. **Alain Rudiger:** Writing – review & editing, Supervision. **Roman Leist:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation. **Lisa Oberli:** Writing – review & editing, Validation, Supervision, Project administration. **Thomas Schweingruber:** Writing – review & editing, Supervision, Resources, Project administration. **Marie-Alice Fraiture:** Writing – review & editing, Validation, Investigation.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.toxrep.2026.102239](https://doi.org/10.1016/j.toxrep.2026.102239).

Data availability

All data has been made available in the manuscript of supplemental information.

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